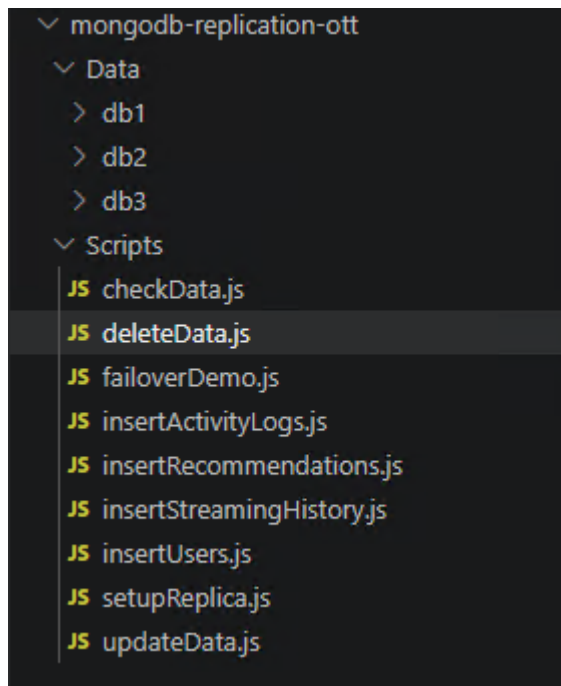


OTT Platform Case Study Assignment – MongoDB Replication & Replica Sets

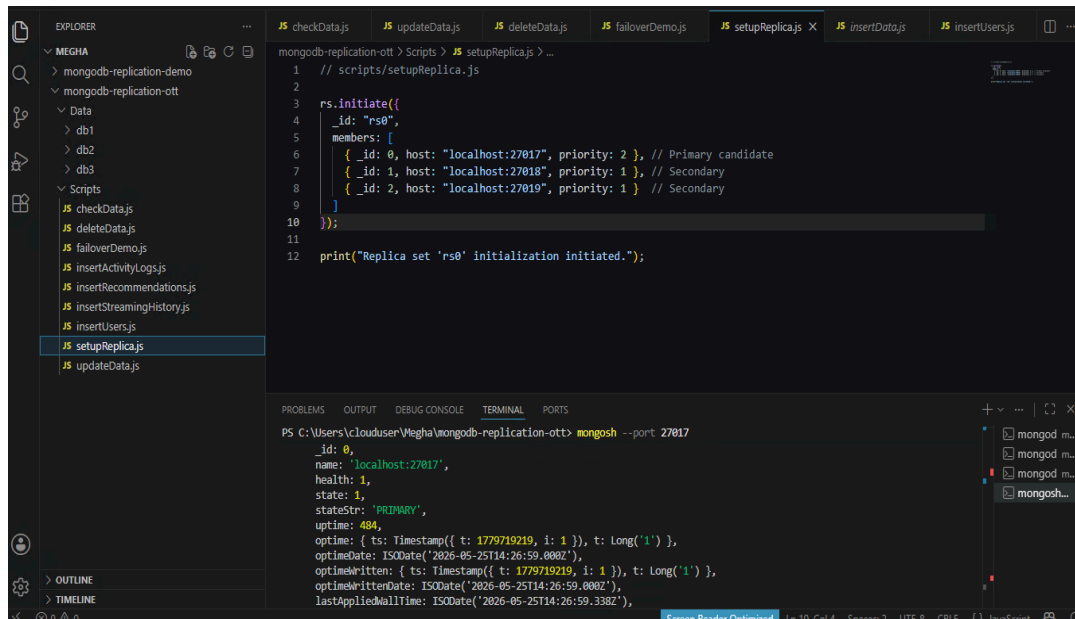


```
mongod --replSet rs0 --port 27017 --dbpath .\data\db1
mongod --replSet rs0 --port 27018 --dbpath .\data\db2
mongod --replSet rs0 --port 27019 --dbpath .\data\db3
```

```
mongosh --port 27017
```

setupReplica.js

```
rs.initiate({
  _id: "rs0",
  members: [
    { _id: 0, host: "localhost:27017" },
    { _id: 1, host: "localhost:27018" },
    { _id: 2, host: "localhost:27019" }
  ]
})
```



use ottPlatform

insertUsers.js

```
db.users.insertMany([
  {
    userId: "U101",
    name: "Rahul",
    email: "rahul@gmail.com",
    subscription: "Premium",
    country: "India",
    createdAt: new Date()
  },
  {
    userId: "U102",
    name: "Anu",
    email: "anu@gmail.com",
    subscription: "Basic",
    country: "India",
    createdAt: new Date()
  }
])
```

print("Users Inserted")

```
mongodb-replication-ott > Scripts > JS insertUsers.js > ...
1 db.users.insertMany([
2   {
3     userId: "U101",
4     name: "Rahul",
5     email: "rahul@gmail.com",
6     subscription: "Premium",
7     country: "India",
8     createdAt: new Date()
9   },
10  {
11    userId: "U102",
12    name: "Anu",
13    email: "anu@gmail.com",
14    subscription: "Basic",
15    country: "India",
16    createdAt: new Date()
17  }
18 ])
19
20 print("Users Inserted")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
P5 C:\Users\clouduser\Megha\mongodb-replication-ott> mongosh --port 27017
rs0 [direct: primary] ottPlatform> db.users.find().pretty()
[
  {
    _id: ObjectId('6a145e9fb40be4173fab114'),
    userId: 'U101',
    name: 'Rahul',
    email: 'rahul@gmail.com',
    subscription: 'Premium',
    country: 'India',
    createdAt: ISODate('2026-05-25T14:37:19.541Z')
  },
  {
    _id: ObjectId('6a145e9fb40be4173fab115'),
    userId: 'U102',
    name: 'Anu',
    email: 'anu@gmail.com',
    subscription: 'Basic',
    country: 'India',
    createdAt: ISODate('2026-05-25T14:37:19.541Z')
  }
]
```

insertStreamingHistory.js

```
db.streaming_history.insertMany([
  {
    historyId: "H101",
    userId: "U101",
    contentTitle: "Stranger Things",
    genre: "Sci-Fi",
    watchDuration: 45,
    watchedAt: new Date()
  },
  {
    historyId: "H102",
    userId: "U102",
    contentTitle: "Money Heist",
    genre: "Thriller",
    watchDuration: 30,
    watchedAt: new Date()
  }
])

print("Streaming History Inserted")
```

```
1 db.streaming_history.insertMany([
2   {
3     historyId: "H101",
4     userId: "U101",
5     contentTitle: "Stranger Things",
6     genre: "Sci-Fi",
7     watchDuration: 45,
8     watchedAt: new Date()
9   },
10  {
11    historyId: "H102",
12    userId: "U102",
13    contentTitle: "Money Heist",
14    genre: "Thriller",
15    watchDuration: 30,
16    watchedAt: new Date()
17  }
18 ])
19
20 print("Streaming History Inserted")
```

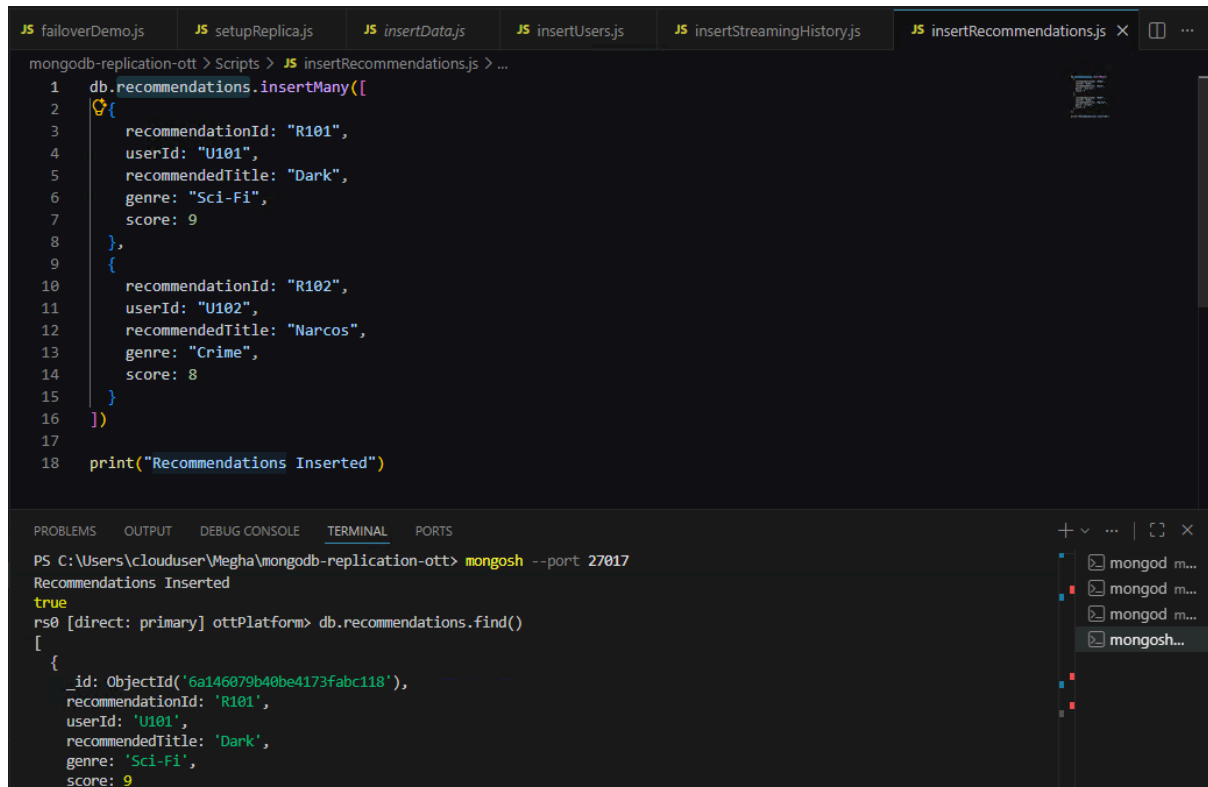
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\clouduser\Megha\mongodb-replication-ott> mongosh --port 27017
Streaming History Inserted
true
rs0 [direct: primary] ottPlatform> db.streaming_history.find()
[
  {
    _id: ObjectId('6a146014b40be4173fab116'),
    historyId: 'H101',
    userId: 'U101',
    contentTitle: 'Stranger Things',
    genre: 'Sci-Fi',
    watchDuration: 45,
```

insertRecommendations.js

```
db.recommendations.insertMany([
  {
    recommendationId: "R101",
    userId: "U101",
    recommendedTitle: "Dark",
    genre: "Sci-Fi",
    score: 9
  },
  {
    recommendationId: "R102",
    userId: "U102",
    recommendedTitle: "Narcos",
    genre: "Crime",
    score: 8
  }
])
```

print("Recommendations Inserted")



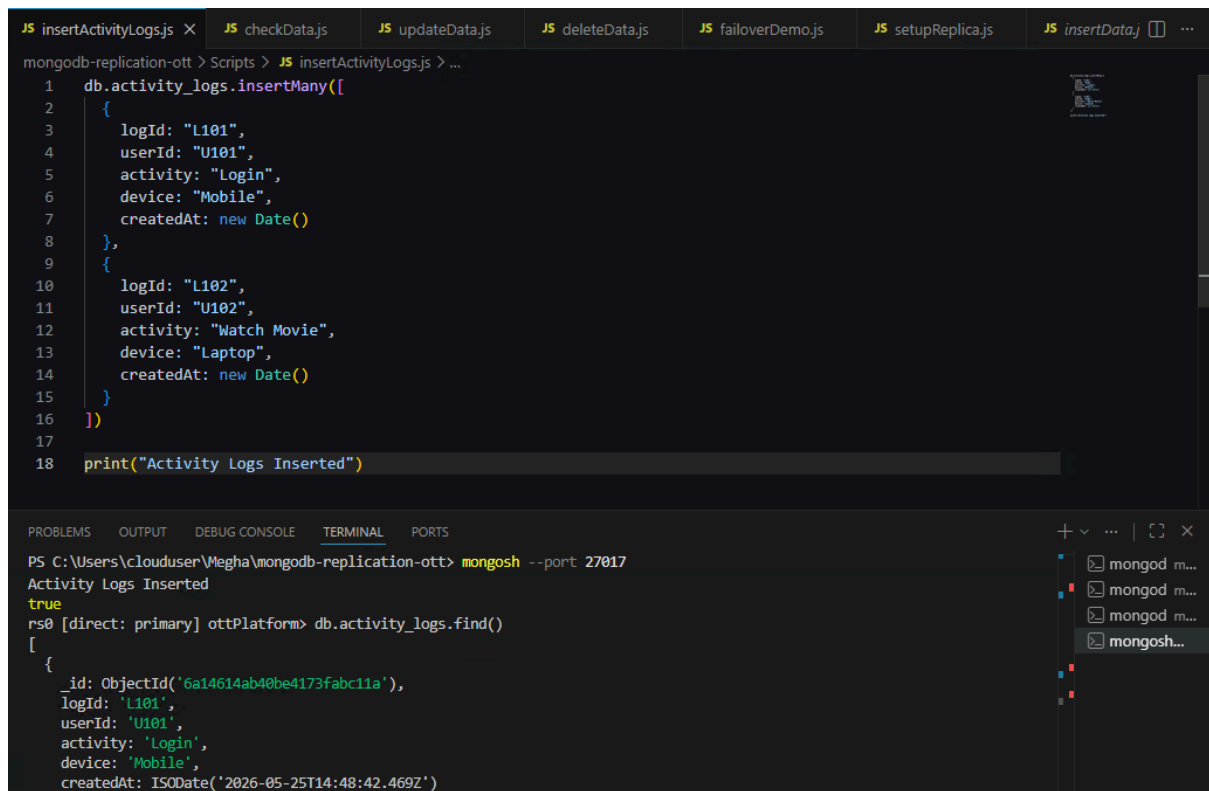
```
JS failoverDemo.js JS setupReplica.js JS insertData.js JS insertUsers.js JS insertStreamingHistory.js JS insertRecommendations.js X ...
mongodb-replication-ott > Scripts > JS insertRecommendations.js > ...
1 db.recommendations.insertMany([
2   {
3     recommendationId: "R101",
4     userId: "U101",
5     recommendedTitle: "Dark",
6     genre: "Sci-Fi",
7     score: 9
8   },
9   {
10    recommendationId: "R102",
11    userId: "U102",
12    recommendedTitle: "Narcos",
13    genre: "Crime",
14    score: 8
15  }
16 ])
17
18 print("Recommendations Inserted")

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\Clouduser\Megha\mongodb-replication-ott> mongosh --port 27017
Recommendations Inserted
true
rs0 [direct: primary] ottPlatform> db.recommendations.find()
[
  {
    _id: ObjectId('6a146079b40be4173fab118'),
    recommendationId: 'R101',
    userId: 'U101',
    recommendedTitle: 'Dark',
    genre: 'Sci-Fi',
    score: 9
  },
  {
    _id: ObjectId('6a146079b40be4173fab119'),
    recommendationId: 'R102',
    userId: 'U102',
    recommendedTitle: 'Narcos',
    genre: 'Crime',
    score: 8
  }
]
```

[insertActivityLogs.js](#)

```
db.activity_logs.insertMany([
  {
    logId: "L101",
    userId: "U101",
    activity: "Login",
    device: "Mobile",
    createdAt: new Date()
  },
  {
    logId: "L102",
    userId: "U102",
    activity: "Watch Movie",
    device: "Laptop",
    createdAt: new Date()
  }
])
```

print("Activity Logs Inserted")



The screenshot shows a VS Code editor with a file named `insertActivityLogs.js` open. The code in the file is as follows:

```
1 db.activity_logs.insertMany([
2   {
3     logId: "L101",
4     userId: "U101",
5     activity: "Login",
6     device: "Mobile",
7     createdAt: new Date()
8   },
9   {
10    logId: "L102",
11    userId: "U102",
12    activity: "Watch Movie",
13    device: "Laptop",
14    createdAt: new Date()
15  }
16 ])
17
18 print("Activity Logs Inserted")
```

The terminal output shows the command `mongosh --port 27017` being executed, followed by the output of the script:

```
Activity Logs Inserted
true
rs0 [direct: primary] ottPlatform> db.activity_logs.find()
[
  {
    _id: ObjectId('6a14614ab40be4173fab11a'),
    logId: 'L101',
    userId: 'U101',
    activity: 'Login',
    device: 'Mobile',
    createdAt: ISODate('2026-05-25T14:48:42.469Z')
  },
  {
    _id: ObjectId('6a14614ab40be4173fab11b'),
    logId: 'L102',
    userId: 'U102',
    activity: 'Watch Movie',
    device: 'Laptop',
    createdAt: ISODate('2026-05-25T14:48:42.469Z')
  }
]
```

[checkData.js](#)

```
print("USERS")
```

```
printjson(db.users.find().toArray())
```

```
print("STREAMING HISTORY")
```

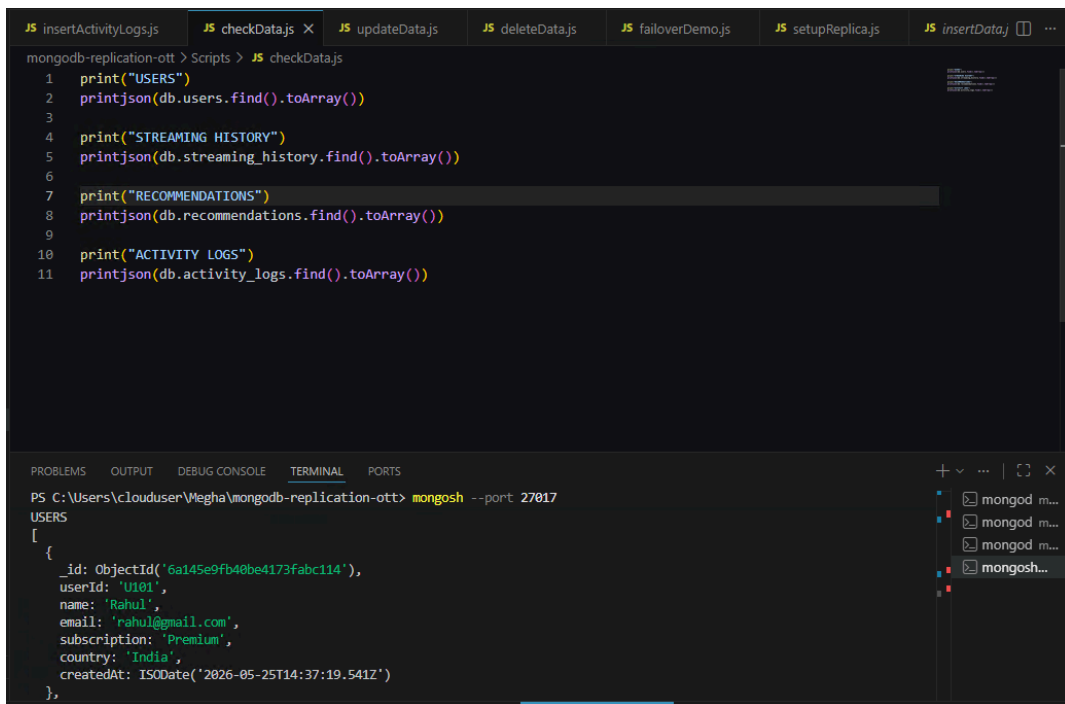
```
printjson(db.streaming_history.find().toArray())
```

```
print("RECOMMENDATIONS")
```

```
printjson(db.recommendations.find().toArray())
```

```
print("ACTIVITY LOGS")
```

```
printjson(db.activity_logs.find().toArray())
```



```
JS insertActivityLogs.js JS checkData.js X JS updateData.js JS deleteData.js JS failoverDemo.js JS setupReplica.js JS insertData.js ...
mongodb-replication-ott > Scripts > JS checkData.js
1 print("USERS")
2 printjson(db.users.find().toArray())
3
4 print("STREAMING HISTORY")
5 printjson(db.streaming_history.find().toArray())
6
7 print("RECOMMENDATIONS")
8 printjson(db.recommendations.find().toArray())
9
10 print("ACTIVITY LOGS")
11 printjson(db.activity_logs.find().toArray())

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\clouduser\Megha\mongodb-replication-ott> mongosh --port 27017
USERS
[
  {
    _id: ObjectId('6a145e9fb40be4173fab114'),
    userId: 'U101',
    name: 'Rahul',
    email: 'rahul@gmail.com',
    subscription: 'Premium',
    country: 'India',
    createdAt: ISODate('2026-05-25T14:37:19.541Z')
  },
]
```

updateData.js

```
db.users.updateOne(
  { userId: "U102" },
  { $set: { subscription: "Premium" } }
)

db.streaming_history.updateOne(
  { historyId: "H102" },
  { $set: { watchDuration: 60 } }
)

db.recommendations.updateOne(
  { recommendationId: "R102" },
  { $set: { score: 10 } }
)

print("Data Updated")
```

```
JS insertActivityLogs.js JS checkData.js JS updateData.js X JS deleteData.js JS failoverDemo.js JS setupReplica.js
mongodb-replication-ott > Scripts > JS updateData.js > ...
1 db.users.updateOne(
2   { userId: "U102" },
3   { $set: { subscription: "Premium" } }
4 )
5
6 db.streaming_history.updateOne(
7   { historyId: "H102" },
8   { $set: { watchDuration: 60 } }
9 )
10
11 db.recommendations.updateOne(
12   { recommendationId: "R102" },
13   { $set: { score: 10 } }
14 )
15
16 print("Data Updated")

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\clouduser\Megha\mongodb-replication-ott> mongosh --port 27017
device: 'Laptop',
createdAt: ISODate('2026-05-25T14:48:42.469Z')
}
]
true
rs0 [direct: primary] ottPlatform>
(To exit, press Ctrl+C again or Ctrl+D or type .exit)
rs0 [direct: primary] ottPlatform>
● PS C:\Users\clouduser\Megha\mongodb-replication-ott> mongosh --port 27017 scripts\updateData.js
Data Updated
```

db.users.deleteOne({ userId: "U102" })

db.streaming_history.deleteOne({ historyId: "H102" })

db.activity_logs.deleteOne({ logId: "L102" })

print("Data Deleted")

```
JS insertActivityLogs.js JS checkData.js JS updateData.js JS deleteData.js X JS failoverDemo.js JS setupReplica.js JS
mongodb-replication-ott > Scripts > JS deleteData.js > ...
1 db.users.deleteOne({ userId: "U102" })
2
3 db.streaming_history.deleteOne({ historyId: "H102" })
4
5 db.activity_logs.deleteOne({ logId: "L102" })
6
7 print("Data Deleted")

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\clouduser\Megha\mongodb-replication-ott> mongosh --port 27017
subscription: 'Basic',
country: 'India',
createdAt: ISODate('2026-05-25T14:37:19.541Z')
}
]
rs0 [direct: primary] ottPlatform>
(To exit, press Ctrl+C again or Ctrl+D or type .exit)
rs0 [direct: primary] ottPlatform>
● PS C:\Users\clouduser\Megha\mongodb-replication-ott> mongosh --port 27017 scripts\deleteData.js
Data Deleted
```